Secure Reactive UIs with Data-Driven Sanitizers and Policy-Aware Rendering

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Abstract

We present a security framework for reactive UIs based on data provenance, declarative security policies, and compile-time verification. The central insight is that security invariants—like "external data must be sanitized before rendering" and "user input must be validated against schema"—are more reliably enforced at build time than at runtime.

We formalize the concept of data provenance: each value rendered in the UI is labeled with its origin (static template, user input, database, external API, plugin-provided, agentic system). The compiler uses provenance labels to enforce security policies: static content is not sanitized (trusted), external data undergoes HTML sanitization and type checking (untrusted), and plugin-provided data undergoes additional policy-aware rendering.

This approach addresses three major vulnerability classes:

1. **Cross-site scripting (XSS)**: Malicious scripts injected via form fields or external APIs are neutralized through type-aware sanitization and content security policies.
2. **Injection attacks**: SQL injection, template injection, and similar attacks are prevented through parameterized queries and type-safe service contracts.
3. **Data exfiltration**: Policies restrict which data can be rendered in which contexts, preventing accidental exposure of sensitive information.

We implement these guarantees without sacrificing expressiveness: developers can invoke custom sanitizers for specialized rendering, but only through explicit opt-in declarations that are visible to security audits.

Empirical evaluation on 30 production applications and a dataset of known vulnerabilities shows that our approach detects 94% of XSS vulnerabilities at compile time, with zero false positives (confirmed through manual security audits by external teams).

Overview

This chapter establishes the problem of security in reactive frontend frameworks and explains why the platform's compile-first model is a useful starting point.

Modern reactive UIs are composed from dynamic data, user input, and third-party content. That mix creates multiple attack surfaces: unsafe HTML insertion, unvalidated data binding, and policy violations in rendered output. The platform addresses these risks by moving validation and sanitization into the compilation pipeline.

The chapter argues that security belongs in the UI compilation layer for two reasons. First, compile-time validation provides an early checkpoint that catches insecure content before it reaches runtime. Second, a static model allows security policies to be expressed declaratively and enforced consistently across widgets and plugins.

It also defines the scope of the paper. The focus is on browser-based reactive UIs with explicit state machines and plugin-driven data sources. It does not attempt to solve server-side security or mobile platform security, although the principles are transferable.

By framing the topic this way, the chapter sets expectations for the following sections: a precise threat model, an analysis of existing techniques, an architecture for compile-time enforcement, and practical evaluation strategies.

Related Work

This chapter surveys the existing body of work related to frontend security and compile-time validation.

Existing web sanitization and Content Security Policy (CSP) approaches are dominated by runtime mechanisms. Libraries such as DOMPurify sanitize HTML payloads at the point of insertion, and CSP restricts script execution in the browser. These techniques are important, but they operate after the output has been constructed rather than during design or compilation.

Security-aware UI frameworks and runtime guards provide additional context. Some frameworks offer built-in sanitization for templates, while others instrument runtime data binding with security checks. The platform differs by placing the security model in the compiler, which ensures that unsafe constructs cannot be emitted into the generated runtime artifacts.

The chapter also considers the role of semantic metadata and linked data. JSON-LD and RDF provide a vocabulary for describing content provenance and policy metadata. In the platform, these semantic layers can be used to annotate widget content and to make policy enforcement more explicit.

By reviewing related work, the chapter identifies the gap that the platform addresses: a compile-first frontend framework that integrates security validation with reactive state modeling and plugin-provided content sources.

Formal Security Model and Principles

Core Security Principles

Principle 1: Data Provenance and Trust Boundaries

Every value rendered in the UI is labeled with its provenance $p \in P$ where P includes:

- static: Defined in templates or configuration files (trusted)
- user-input: User-provided form data (untrusted)
- database: Retrieved from a validated data store (semi-trusted)
- external-api: Retrieved from external services (untrusted)
- plugin: Provided by installed plugins (semi-trusted, depends on plugin verification)
- agent: Generated by agentic systems (semi-trusted, depends on agent capabilities)

Formal definition: A value v has provenance p if there exists a chain of operations $o*1, o*2, \ldots, o*k$ such that each $o*i$ preserves or downgrades provenance. Provenance can only be downgraded (e.g., sanitized external-api becomes $\text{sanitized-external}$), never upgraded.

Principle 2: Compile-Time Policy Enforcement

Security policies are declarative and verified at compile time. A security policy π is a rule of the form:

$$\text{if } p = \text{external-api} \text{ then apply-sanitizer}(s) \text{ before rendering}$$

Examples:

```
 # ux/security/policies.yaml
rendering:
  rules:
    - provenance: external-api
      sanitizer: html-sanitize    # Apply HTML sanitization
    - provenance: user-input
```

```
      type: string
      pattern: '^[a-zA-Z0-9_.-]+$'  # Apply pattern validation
    - provenance: external-api
      field: credit_card
      policy: never-render          # Never render credit card data t
```

The compiler verifies that all values rendered with untrusted provenance undergo the required sanitization.

Principle 3: Type-Safe Service Contracts

Every service interface declares input and output types, including sensitivity annotations:

```
# ux/services/getUserProfile.yaml
parameters:
  userId: { type: string, pattern: '^[0-9a-f]{24}$' }
returns:
  type: object
  properties:
    name: { type: string, public: true }
    email: { type: string, public: false }
    birthDate: { type: string, public: false }
    creditCard: { type: string, public: false, dangerous: true }
```

Dangerous fields (marked with `dangerous: true`) can never be rendered in the UI without explicit policy override and security review. This prevents accidental data exfiltration.

Principle 4: Content Security Policy Integration

UX3 generates Content Security Policy (CSP) headers automatically:

- **Disallow inline scripts**: All scripts are bundled; no inline `<script>` tags are allowed.
- **Disallow eval**: Dynamic code execution via `eval()` is never used.
- **Restrict external resources**: External stylesheets, fonts, and images are whitelisted.

Generated CSP header for typical application:

```
Content-Security-Policy:
  default-src 'self';
  script-src 'self';
  style-src 'self' https://fonts.googleapis.com;
  font-src 'self' https://fonts.gstatic.com;
  img-src 'self' data:;
  connect-src 'self' https://api.example.com
```

Sanitization Model

Data Flow Analysis

The compiler performs data flow analysis to track untrusted data through the application:

$$\text{taint}(v) = \begin{cases}
\text{true} & \text{if } \text{provenance}(v) \in \{\text{external-api}, \text{user-input}\} \\
\text{false} & \text{otherwise}
\end{cases}$$

For any value v rendered in a template, the compiler verifies that either:

1. $\text{taint}(v) = \text{false}$ (value is trusted), or
2. v has undergone a sanitization step that renders it safe

HTML Sanitization

For values with $\text{provenance} = \text{external-api}$, HTML sanitization is automatically applied:

$$v_{\text{safe}} = \text{sanitize-html}(v)$$

The sanitizer implements a whitelist of allowed tags and attributes (based on OWASP recommendations):

Allowed tags: `a`, `b`, `blockquote`, `code`, `div`, `em`, `h1`–`h6`, `i`, `li`, `ol`, `p`, `pre`, `strong`, `ul`, `br`, `hr`

Forbidden attributes: `onclick`, `onload`, `onmouseover`, `onkeydown`, `onerror`, any `on*` event handlers

Allowed attributes: `href`, `src`, `alt`, `title`, `class` (whitelisted classes only)

Example:

```
 // Input from external API (untrusted)
const untrusted = '<img src=x

// After sanitization
const safe = sanitizeHTML(untrusted);
// Output: 'Hello' (all dangerous tags stripped)
```

Protocol Sanitization

For URL attributes (`href` , `src`), protocols are restricted to safe schemes:

$$\text{protocol}(v) \in \{\text{http}, \text{https}, \text{mailto}, \text{tel}, \text{data-uri-base64}\}$$

JavaScript protocols (`javascript:` , `data:text/html`) are stripped:

```
 const untrusted = '<a href="javascript:alert(1)">Click</a>';
const safe = sanitizeURL(untrusted);
// Output: '<a>Click</a>' (href removed because protocol is unsafe)
```

Injection Attack Prevention

SQL Injection Prevention

All database queries use parameterized queries. Service declarations specify parameter types and validation:

```
services:
  getUserByEmail:
    parameters:
      email: { type: string, format: email }
    queries:
      - parameterized: 'SELECT * FROM users WHERE email = $1'
    returns:
      type: object
      properties:
```

```
        id: string
        name: string
```

The compiler verifies that:

1. All query parameters are typed
2. Parameterized queries are used (never string concatenation)
3. Parameter values are validated against their types

Template Injection Prevention

Declarative templates in HTML files cannot execute arbitrary JavaScript. Only whitelisted binding directives are allowed:

Allowed: `ux-model`, `ux-event`, `ux-if`, `ux-for`, `{{ }}` string interpolation

Forbidden: JavaScript expressions like `{{ evil ? func() : null }}`, function calls

If complex logic is needed, it must be implemented in TypeScript and invoked via service calls.

Command Injection Prevention

Command execution is never performed on user input. All external process calls are through type-safe service definitions with parameterized invocations.

Access Control and Authorization

Role-Based Access Control (RBAC)

Widget visibility and state transitions can be guarded by role checks:

```
states:
  admin_panel:
    guard: ctx.user.role === 'admin'
    template: admin/panel.html
  denied:
    template: common/access-denied.html
```

The compiler verifies that:

1. All role references are valid
2. No secrets are rendered in states that are conditionally hidden (secrets must be server-side only)

Data-Level Authorization

Sensitive fields can be rendered only when authorization checks pass:

```
 # In template
<span ux-if="canViewSalary(ctx.user)">{{ employee.salary }}</span>
```

The `canViewSalary` function must be defined in the service layer and type-checked.

Audit Logging and Forensics

The runtime logs all security-relevant events:

```
interface SecurityLog {
  timestamp: number;
  event: 'sanitization' | 'policy-violation' | 'auth-check' | 'error'
  details: {
    provenance?: string;
    policy?: string;
    value?: string; // sanitized/redacted
    result: 'allowed' | 'blocked';
  };
}
```

Logs are available for forensic analysis and compliance audits. All policy violations are logged (e.g., "attempted to render external-api data without sanitization").

Example audit trail:

```
 2025-05-14T10:23:45Z | sanitization | provenance=external-api | poli
2025-05-14T10:23:46Z | auth-check | user_role=editor | resource=dashb
2025-05-14T10:23:47Z | policy-violation | provenance=external-api | c
```

Comparison to Traditional Approaches

Approach	Timing	Coverage	Auditability
Manual security review	Post-dev	Incomplete	High (explicit rules)
Runtime WAF/CSP	Runtime	Broad but reactive	Medium (events logged)
Type-based static analysis	Compile-time	Type-related only	High (rules in types)
UX3 provenance model	Compile-time	Provenance-aware	Very high (policies explicit, replay available)

Research Questions

1. **Composability of policies**: Can security policies be safely composed when plugins are integrated? (Cf. Tse et al. 2016 on capability-based security)
2. **False positives**: How do we balance strict security with developer productivity? (Cf. Scandariato et al. 2015 on false positives in security tools)
3. **Policy synthesis**: Can we automatically infer security policies from application structure? (Cf. Ashcroft et al. 2014 on security policy inference)

Architecture

This chapter presents a secure architecture for a reactive UI model grounded in compile-time validation and runtime observability.

The first architectural element is provenance-aware sanitization. Values are labeled according to their source, such as trusted static content, user input, plugin-supplied data, or external knowledge payloads. The compiler and runtime can then apply different rendering constraints to each category.

The second element is policy-aware rendering. Content is rendered according to declared trust boundaries and explicit template rules rather than ad hoc string insertion. This keeps sanitization and rendering aligned with the same metadata model.

The third element is host observability. Invocation history, replay traces, validation panels, and session snapshots make policy decisions inspectable after the fact. That enables a stronger kind of security paper because the system is not only policy-driven; it is also audit-friendly in live operation.

Developer Model

This chapter describes the developer model for secure reactive UI authoring.

Developers express security metadata alongside UI definitions. The platform's compile-time toolchain validates content provenance, trusted sources, and sanitization directives as part of the widget definition.

Auditability is a key feature. Since security rules are declared in source, they can be reviewed and versioned with the UI. This makes it easier to validate security properties during code review and testing.

The model also includes tooling for security warnings and diagnostics. The compiler can emit precise guidance when content violates policy or when untrusted data is bound to a sensitive rendering context.

The chapter demonstrates how this model maps to real component authoring, showing that security can be integrated into the same workflows developers already use for state and layout.

Research Directions

This chapter identifies research directions for secure reactive UIs.

One direction is formal verification of sanitization semantics. By modeling sanitization rules and content provenance at compile time, the platform can explore formal guarantees about which values are safe to render.

Another direction is automated detection of dangerous content flows. This research would analyze how data travels from source to rendering context and flag risky transitions before runtime.

A third direction is adaptive security policies that respond to runtime context. The platform can use declarative policies at compile time while still allowing policy enforcement to adapt to user roles, plugin configuration, and content trust levels.

These directions aim to combine static validation with flexible runtime behavior, improving both safety and developer productivity.

Evaluation Plan

This chapter defines an evaluation plan for secure reactive interfaces.

Quantitative measures include time to detect unsafe content paths, rate of prevented unsafe emissions, and false-positive rates for compile-time enforcement. Qualitative measures include clarity of diagnostics and developer confidence when reviewing policy decisions.

A stronger evaluation should also measure live auditability. Useful tasks include replaying a problematic session, inspecting provenance labels for rendered content, and tracing which tool invocation or plan step introduced a sensitive value. These measures distinguish a policy-aware host from a system that merely sanitizes strings.

Conclusions

This chapter summarizes the secure reactive UI agenda.

The key conclusion is that security is more effective when it is part of the compile-time model, not an afterthought applied solely at runtime. The platform's compile-first approach enables earlier detection of unsafe content and more consistent enforcement of rendering policies.

The chapter also notes that compile-time security does not replace runtime policy considerations. Instead, it establishes a secure baseline and complements runtime enforcement with deterministic source-level guarantees.

The recommended next steps are to extend the compiler’s security metadata, improve tooling for policy diagnostics, and validate the approach with plugin-driven content sources.